Abstract

Rising energy prices not only increase household living costs but may also lead to different spending adjustments across regions with different levels of petroleum dependence. Understanding these differences can help reveal regional economic pressures that may be hidden by changes in national average spending. Previous studies have found that the effects of oil price changes on consumption vary depending on an economy's exposure to oil (De Michelis et al., 2019). Changes in gasoline prices can also affect household spending (Gelman et al., 2023). However, it remains unclear whether non-durable consumer spending across US states changed differently according to their pre-war petroleum consumption levels during the energy price shock following the 2022 Russia–Ukraine War. Identifying this relationship is challenging because all states experienced rising oil prices while also being affected by other economic factors, such as the recovery from the COVID-19 pandemic. In addition, changes in nominal spending reflect both price changes and adjustments in the quantities purchased.

To examine this question, this study combines state-level Affinity non-durable spending data with average petroleum consumption per capita from 2019 to 2021. The dataset covers 50 US states, including 27 weeks before and 27 weeks after the outbreak of the war. We use a continuous-exposure difference-in-differences (DID) model with state and week fixed effects, and cluster standard errors at the state level. A Stute nonparametric test is used to examine pre-war trends. We also conduct sensitivity analyses by using an alternative year to measure exposure and adding regional gasoline prices as a control variable.

The results show that a one-standard-deviation increase in pre-war petroleum exposure is associated with a 0.415 percentage point lower change in the non-durable spending indicator after the war relative to the pre-war period. However, the statistical evidence is limited ($p = 0.0523$). The estimated effects remain generally similar in direction and size across the additional model specifications. By using continuous differences in petroleum exposure across states, this study provides empirical evidence on regional spending adjustments during an energy price shock. However, the estimated relationship should not be directly interpreted as a causal effect of petroleum dependence itself.

1.Introduction

Russia’s full-scale invasion of Ukraine in February 2022 created substantial uncertainty in global energy markets. The U.S. Energy Information Administration (EIA) reports that geopolitical tensions surrounding Russia, together with already low global oil inventories, contributed to the sharp increase in crude oil prices during the first half of 2022 (EIA, 2023). The invasion was an important part of the 2022 oil-price episode, alongside the post-pandemic recovery in demand and existing supply constraints. Higher crude oil prices can raise the energy costs faced by households and businesses. If these costs are passed through to consumers while household income does not rise at the same pace, purchasing power available for other spending may fall.

The effects of this shock may differ across U.S. states. States differ considerably in how intensively petroleum products are used. Transport patterns, population density, climate and industrial activity can all contribute to differences in petroleum consumption per resident. A common increase in energy prices may therefore create different spending pressures across states.

This motivates our main research question: following the oil-price shock associated with Russia's invasion of Ukraine, did states with higher pre-war petroleum consumption per capita experience weaker growth, or larger relative declines, in nominal non-durable consumer spending than states with lower pre-war petroleum consumption per capita?

This study employs a continuous-exposure difference-in-differences (DID) framework, using average per capita petroleum consumption over 2019–2021 to measure states' pre-war petroleum-use intensity. Controlling for state and week fixed effects, we examine whether changes in the nominal non-durable consumer spending index following the invasion vary systematically with pre-war petroleum exposure.

2. Related Literature and Economic Mechanisms

2.1 Oil Prices and Consumer Spending

Existing research shows that energy prices can affect consumer spending, although the effects depend on the source of the shock. Using high-frequency transaction data, Gelman et al. (2023) find that gasoline-price changes affect consumer spending, suggesting that their effects extend beyond fuel expenditure. Kilian (2009) emphasizes that oil shocks originating from different sources have different economic consequences. The 2022 oil-price increase should therefore be considered alongside the post-pandemic recovery in demand and existing supply constraints, rather than attributed entirely to Russia's invasion of Ukraine. This study examines nominal non-durable spending, which reflects changes in both prices and quantities purchased.

2.2 Petroleum Exposure and Regional Heterogeneity

National averages may conceal differences in regional responses to oil-price shocks. De Michelis et al. (2019) show that consumption responses differ across countries and U.S. states according to their oil exposure. Their state-level analysis uses a different exposure measure and new car registrations as a consumption indicator, leaving scope to examine other spending categories and shock episodes.

Building on this literature, our study examines whether nominal non-durable spending adjustments during the 2022 energy-price episode varied with states' pre-war petroleum-use intensity. It extends the analysis of regional heterogeneity to frequently purchased goods and a specific geopolitical shock.

2.3 Transmission Channels and Research Hypothesis

Russia's invasion of Ukraine added supply uncertainty to an already tight oil market, contributing to rising crude oil prices in early 2022 (EIA, 2023). Higher fuel prices can increase household transport expenses (EIA, 2014), while higher production and logistics costs may be passed on to

consumers. If nominal income does not keep pace, households may reduce purchases or switch to cheaper alternatives. These pressures may be greater in states with higher petroleum-use intensity.

However, competing effects make the net relationship uncertain. Higher gasoline prices may directly increase recorded nominal spending, while petroleum-producing states may benefit from higher industry income and employment. These effects could partly offset the pressure on household budgets.

We therefore test the hypothesis that, following Russia's invasion of Ukraine, states with higher pre-war petroleum exposure experienced smaller increases, or larger declines, in nominal non-durable spending relative to the pre-war period.

3. Data and Variable Construction

3.1 Data Sources and Sample Construction

This study combines state-level Affinity non-durable spending data from Opportunity Insights, petroleum consumption per capita data from the U.S. Energy Information Administration (EIA), weekly West Texas Intermediate (WTI) crude oil prices, and regional retail gasoline prices to construct a state–week panel dataset for the empirical analysis. The non-durable spending indicator is used as the dependent variable, while petroleum consumption per capita is used to construct the continuous pre-war exposure measure. Weekly WTI crude oil prices are used to examine oil price movements after the outbreak of the war and provide supporting evidence for the choice of the post-war observation window. Regional retail gasoline prices are used in additional analysis. All data processing and model estimation are conducted in R.

We match the spending and exposure data using state-level FIPS codes. The reported observation dates of the spending indicator are assigned to weeks ending on Sunday, and the available indicator values are averaged within each state and week to construct the state–week panel. The original data contain both daily and weekly reporting frequencies. Therefore, a week with only one reported observation does not necessarily mean that transaction data for the remaining days are missing. Since the spending indicator is already calculated using a seven-day moving average, the constructed weekly data represent the average of the published indicator values within each week, rather than total weekly consumer spending.

The observation window includes 27 pre-war weeks and 27 post-war weeks, covering August 22, 2021, to February 20, 2022, and March 6, 2022, to September 4, 2022, respectively. All dates refer to the end of each week. Russia's full-scale invasion of Ukraine began on February 24, 2022. The week ending February 27 contains both pre-war and post-war days, so this transition week is excluded from the analysis. The week ending March 6 is treated as the first full post-war week. The same observation window is used for the main regression and additional analyses.

The estimation sample includes states with valid exposure and spending data. If all 50 states have valid observations for the full 54 weeks, the panel contains 2,700 state–week observations. The final sample size depends on the observations included in the actual regressions. Regional gasoline prices are matched to states based on their Petroleum Administration for Defense Districts (PADD) regions. States within the same PADD region are assigned the same weekly regional gasoline price.

3.2 Nominal Non-durable Consumer Spending

The dependent variable is the non-durable spending indicator (spend_nondurables) from the Affinity dataset. It covers several spending categories, including clothing and accessories (CLO), food and beverage stores (FBS), gas stations (GAS), general merchandise (GEN), health and personal care stores (HPC), MSC categories, and wholesale trade (WHO).

This indicator measures changes in spending relative to the baseline period from January 6 to February 2, 2020. It is seasonally adjusted and calculated using a seven-day moving average. Therefore, it does not represent the actual amount of spending in dollars or the week-on-week growth rate. The analysis of this indicator reflects changes in nominal spending relative to the baseline period and should not be directly interpreted as changes in real consumption quantities or household welfare.

In addition, the indicator includes spending at gas stations. Therefore, rising energy prices may affect this indicator in two ways. On the one hand, higher energy costs may reduce household spending on other goods. On the other hand, higher fuel prices may directly increase the amount spent on gasoline, even if the quantity purchased does not increase. Changes in the non-durable spending indicator should therefore be interpreted with these spending categories in mind.

3.3 Pre-war Petroleum Exposure

Pre-war petroleum exposure is measured using the average petroleum products consumption per capita for each state from 2019 to 2021, expressed in barrels per person per year:

$$Exposure_i = \frac{Petroleum_{i,2019} + Petroleum_{i,2020} + Petroleum_{i,2021}}{3}$$

where PC_i, y represents petroleum products consumption per capita in state i in year y. This measure is used as a proxy for each state's reliance on petroleum consumption. It is constructed entirely from data collected before the outbreak of the war and remains constant throughout the study period.

Using a three-year average reduces reliance on data from a single year. However, petroleum consumption in 2020 and 2021 may still have been affected by the COVID-19 pandemic. The main analysis keeps exposure as a continuous variable rather than dividing states into high- and low-exposure groups. This allows us to examine the relationship between pre-war petroleum exposure and relative changes in non-durable consumer spending after the war.

For the sensitivity analysis, we use petroleum consumption per capita in 2019, before the COVID-19 pandemic, as an alternative exposure measure. The alternative measure uses the same scaling as the main model. We also keep the same state–week sample and observation window to make the estimates more comparable across different exposure definitions.

This analysis helps us examine whether the results are sensitive to the choice of years used to measure exposure. However, it cannot completely rule out the effects of the COVID-19 pandemic on consumer spending trends.

3.4 Descriptive Statistics and Spending Patterns

Variable	Unit	Level	N	Mean	SD	Min	Median	Max
<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Non-durable spending: full sample	% change from baseline	State-week	2700	19.4	8.21	-8.99	19.6	54
Non-durable spending: pre-war	% change from baseline	State-week	1350	16.2	7.37	-8.99	16.1	40.6
Non-durable spending: post-war	% change from baseline	State-week	1350	22.6	7.76	-4.82	22.5	54
Petroleum exposure: 2019-2021 average	Barrels per capita per year	State	50	23.4	13.1	12.2	19.7	87.2
Standardized petroleum exposure	Standard deviations	State	50	0	1	-0.863	-0.285	4.88

Table 1 Descriptive Statistics of Main Variables

Table 1 presents the descriptive statistics of the main variables. The sample includes 2,700 state–week observations from 50 US states, with 1,350 observations in both the pre-war and post-war periods. The mean nominal non-durable spending indicator increased from 16.2% before the war to 22.6% after the war, while the median increased from 16.1% to 22.5%. This shows that the spending indicator was generally higher in the post-war sample than in the pre-war sample. These values represent changes in spending relative to the 2020 baseline period. The difference between the two-period means is approximately 6.4 percentage points, but this should not be interpreted as a causal effect of the war.

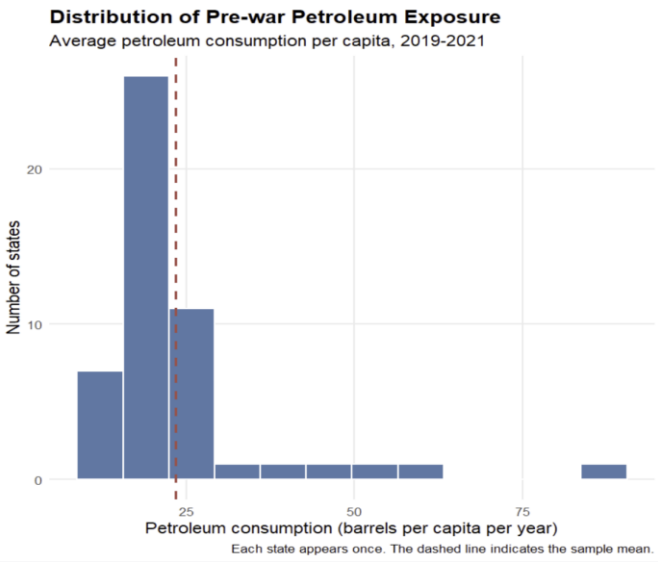


Figure 2. Distribution of Pre-war Petroleum Exposure Across U.S. States Exposure Across U.S. States

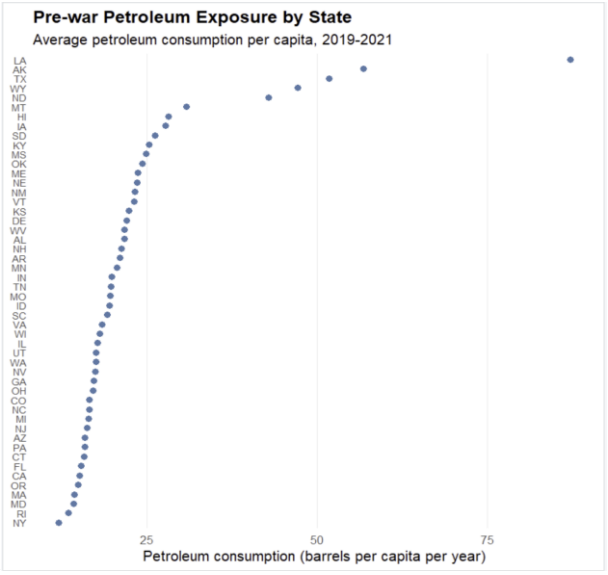


Figure 3. Pre-war Petroleum Exposure by U.S. State

Figures 2 and 3 show differences in pre-war petroleum exposure across US states. Figure 2 presents the overall distribution, while Figure 3 shows the exposure levels of individual states. Overall, the distribution is clearly right skewed. Most states have relatively low or moderate exposure levels, while a small number of states have much higher exposure, creating a long right tail. The sample mean is 23.4 barrels per person per year, which is higher than the median of 19.7, consistent with this distribution. The state-level ranking further shows that Louisiana (LA), Alaska (AK), Texas (TX), Wyoming (WY) and North Dakota (ND) are among the states with the highest exposure levels. Louisiana has the highest exposure at 87.2 barrels per person per year, which is much higher than most other states. In contrast, New York (NY) has the lowest exposure at 12.2 barrels per person per year. These results show that petroleum exposure is not evenly distributed across states.

Instead, most states are relatively concentrated at lower levels, while a few states have much higher exposure.

These differences provide the variation in petroleum exposure needed for the continuous exposure DID analysis. However, they also suggest that the estimated results may be sensitive to a small number of highly exposed states, particularly Louisiana. In addition, this measure includes petroleum consumption by households, businesses and other sectors. Therefore, it should be understood as a measure of overall state-level petroleum-use intensity rather than a direct measure of household fuel expenditure.

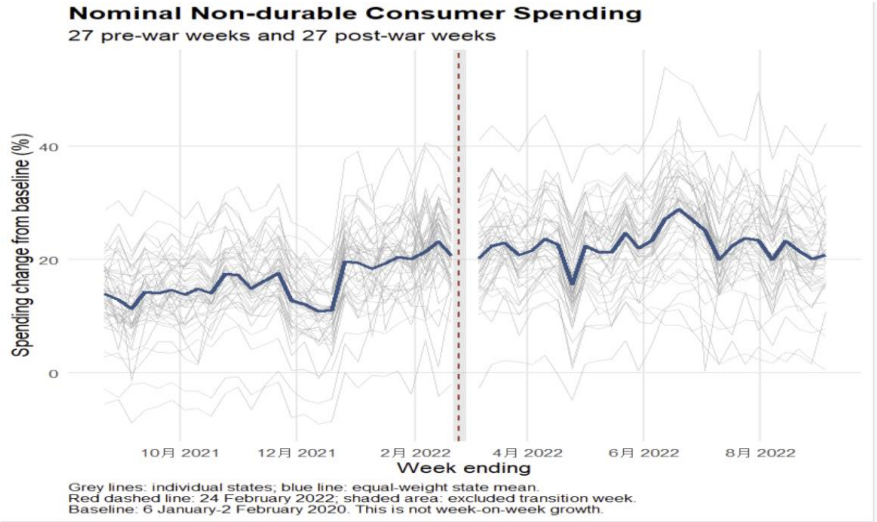


Figure 4. Nominal Non-durable Consumer Spending Before and After the Russia–Ukraine War

Figure 4 shows the changes in nominal non-durable consumer spending over the study period. The graph does not show a continuous overall decline in spending after the war. The blue line, which represents the equally weighted mean across states, shows that spending had already fluctuated before the war. The spending indicator declined towards the end of 2021 before recovering in the weeks leading up to the invasion. After the war began, the average spending indicator continued to fluctuate, reaching a local peak around mid-2022 before declining again. By the end of the observation period, it was close to 20%. Therefore, the graph does not show a sustained or widespread decline in nominal spending following the outbreak of the war. The grey lines, representing individual states, also show considerable differences in spending levels and changes across states.

4. Empirical Strategy

4.1 Oil-price Dynamics and the Post-war Observation Window

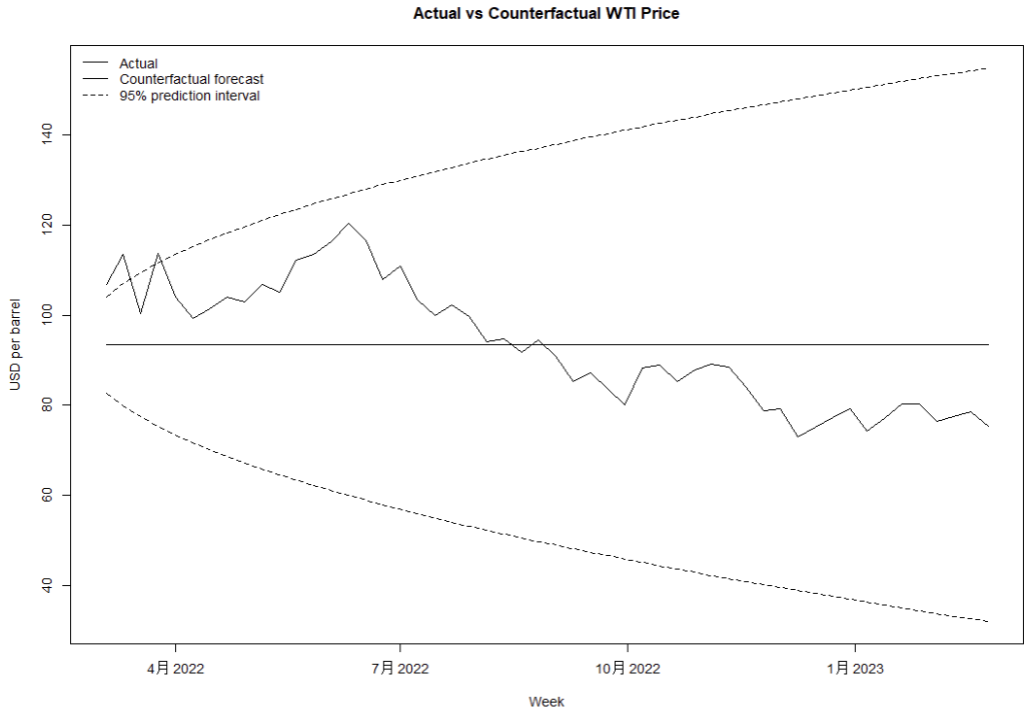
Weekly West Texas Intermediate (WTI) crude oil prices provide descriptive evidence on the timing of the energy price shock. Using the forecast package in R, we compare automatically selected ARIMA models with a random walk benchmark. Forecast performance is evaluated using a pre-war holdout sample and rolling-origin validation. Root mean squared error (RMSE) and mean absolute error (MAE) are used to measure forecast accuracy. The reported results support the selection of the ARIMA (1,1,0) model. We also examine the model residuals to check whether substantial serial correlation remains after fitting the model.

The selected model is fitted using pre-war oil price data and then used to forecast prices after the outbreak of the war. Actual oil prices were initially higher than the predicted path and gradually moved closer to it. This pattern provides supporting evidence for choosing the first 27 complete post-war weeks as the observation window.

The observed oil price pattern is broadly consistent with Zhang et al. (2024), who find that the Russia–Ukraine War and related events contributed to higher crude oil prices. However, the two studies use different methods and observation windows, so their findings cannot be treated as direct validation of our 27-week window.

It is important to note that the forecast is only a statistical benchmark based on historical oil price movements. It cannot determine what oil prices would actually have been if the war had not occurred. Other changes in oil supply and demand, together with forecasting uncertainty, may also explain differences between actual and predicted prices.

Zhang et al. (2024) also suggest that the war may have changed the longer-term behavior of crude oil prices. This highlights the possibility of structural changes when forecasting future prices based on pre-war relationships. Therefore, our 27-week post-war observation window is used to examine short-term spending changes during this period, rather than to identify the exact duration of the war's causal effects.



Figurer 7. Observed and ARIMA-Forecast WTI Crude Oil Prices

4.2 Continuous Difference-in-Differences Specification

This study uses a continuous-exposure difference-in-differences (DID) model to examine whether changes in nominal non-durable consumer spending across US states after the Russia–Ukraine War are systematically related to pre-war petroleum-use intensity.

We measure pre-war exposure using each state's average petroleum products consumption per capita from 2019 to 2021. The exposure measure is standardized across states as follows

$$E_i^* = \frac{E_i - \overline{E}}{s_E}$$

where E_i represents average pre-war petroleum products consumption per capita in state i, while $\bar{E}$ and s_E represent its mean and standard deviation across states. The measure remains constant throughout the study period. Standardization preserves the continuous differences in exposure across states and allows us to interpret the estimated coefficient in terms of a one-standard-deviation increase in exposure.

The baseline model is specified as:

$$Y_{it} = \alpha_i + \lambda_t + \beta(E_i^* \times Post_t) + \varepsilon_{it}.$$

where Y_{it} is the Affinity non-durable spending indicator for state i in week t, calculated as the average of the available published indicator values for that state and week. This indicator measures changes in nominal spending relative to the 2020 baseline period, rather than total spending in dollars or week-on-week growth.

$Post_t$ is an indicator variable equal to 1 during the post-war period and 0 during the pre-war period. The estimation window includes 27 pre-war weeks and 27 post-war weeks, excluding the transition week that contains both pre-war and post-war dates.

State fixed effects (α_i) control for time-invariant characteristics of each state, while week fixed effects (λ_t) capture weekly changes that are common to all states. Since standardized exposure does not change over time within a state, its separate effect is absorbed by the state fixed effects. Similarly, the post-war indicator varies only over time, so its separate effect is absorbed by the week fixed effects. Therefore, these two variables are not estimated separately. Instead, their interaction is used to estimate relative changes in spending across states with different exposure levels.

The main coefficient, β, measures the difference in the average post-war spending change relative to the pre-war period associated with a one-standard-deviation increase in pre-war petroleum exposure. Since the regression uses the original spending indicator expressed as a proportion rather than multiplying it by 100, 100β represents the corresponding difference in percentage points.

If β is negative, states with higher exposure experienced relatively smaller increases or larger declines in the spending indicator after the war. This does not mean their spending must have fallen in absolute terms. Even if spending increased in all states, those with higher exposure may still have experienced weaker growth.

The model uses differences in pre-war petroleum exposure across states together with changes over time before and after the invasion. Since week fixed effects absorb common national changes, β captures differences in spending adjustments associated with pre-war petroleum exposure rather than the overall effect of the war on US consumer spending.

A causal interpretation of this coefficient still depends on the identification assumptions discussed in the next section. These include parallel spending trends across different exposure levels in the

absence of the shock, as well as the absence of other simultaneous shocks that are related to exposure and affect spending.

4.3 Identification Assumptions

The continuous-exposure DID model relies on counterfactual parallel trends: in the absence of the shock, expected changes in non-durable spending would not systematically vary with pre-war petroleum exposure. This allows states to have different spending levels but requires their counterfactual spending changes to be comparable across exposure levels. A causal interpretation also requires no differential anticipation effects and no simultaneous shocks associated with exposure that independently affect spending. State and week fixed effects remove time-invariant state characteristics and common weekly shocks, but do not guarantee these conditions.

Even under parallel trends, the estimated slope does not necessarily identify the causal effect of increasing petroleum exposure itself. States may differ in industrial structure, transport options and other characteristics that influence their responses to the shock. To illustrate, suppose the shock effect at state i 's observed exposure is:

$$\tau_i(E_i^*) = \theta E_i^* + r_i,$$

where $\theta$is an assumed common linear exposure effect and r_i captures remaining differences in shock responses. Under counterfactual parallel trends, the population DID slope is:

$$\beta = \theta + \frac{\mathrm{Cov}(E_i^*, r_i)}{\mathrm{Var}(E_i^*)}.$$

Thus, interpreting β as θ additionally requires $\mathrm{Cov}(E_i^*, r_i) = 0$. Otherwise, the slope reflects both exposure intensity and other exposure-related differences in responses to the shock.

To examine whether observed pre-war trends are consistent with the identification assumption, we use a zero-order Stute nonparametric test. The analysis includes states with complete observations for all 27 pre-war weeks. Taking the week ending February 20, 2022, as the reference week t_0, we define:

$$\Delta Y_{it} = Y_{it} - Y_{it_0}, \qquad t \in \mathcal{T}_{pre} \setminus \{t_0\}.$$

The joint null hypothesis is:

$$H_0: \mathbb{E}[\Delta Y_{it} \mid E_i^* = e] = c_t \quad \text{for all pre-war comparison weeks and exposure levels } e,$$

where c_t may vary across weeks but does not depend on exposure. The reference week is excluded because its change is zero by construction. This test examines whether expected spending changes are related to exposure without restricting that relationship to be linear.

Rejection would indicate exposure-related differences in observed pre-war trends, raising concerns about the DID specification. Failure to reject indicates insufficient evidence of such differences, but cannot establish that counterfactual parallel trends would hold after the war. The test results are reported in Section 5.1.

5. Empirical Results

5.1 pre-war war Trend Diagnostics

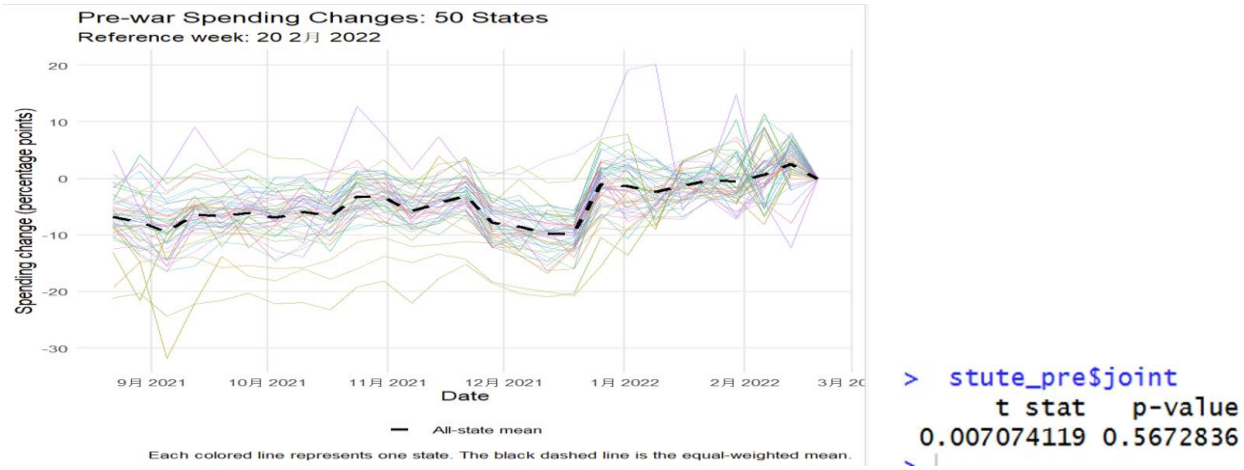


Figure 8. Pre-war Changes in Non-durable Spending Across U.S. States

Figure 8 shows changes in the non-durable spending indicator across 50 U.S. states during the pre-war period. Each colored line represents one state, while the black dashed line shows the equally weighted average of spending changes across all states.

It is important to note that all state-level lines reach zero in the reference week because of how the spending changes are calculated. This does not provide evidence that the parallel trends assumption holds. Similarly, differences and fluctuations among the state-level lines do not necessarily mean that parallel trends are violated. The key question is whether these changes are systematically related to pre-war petroleum exposure.

The pre-war Stute joint test produces a test statistic of 0.00707 and a p-value of 0.5673. Therefore, we fail to reject the null hypothesis at both the 5% and 10% significance levels. This suggests that, under the current sample and test specification, there is insufficient statistical evidence that expected spending changes relative to the reference week systematically vary with petroleum exposure during the pre-war period.

The result is consistent with the comparability of pre-war trends required for the continuous exposure DID analysis and provides some support for the following estimation.

However, failing to reject the null hypothesis does not prove that counterfactual parallel trends hold. The test only examines observed pre-war changes and cannot determine whether states would have followed comparable trends after the war in the absence of the shock. It also cannot rule out differences that may not be detected because of sample size or limited statistical power. In addition, the pre-war test cannot eliminate concerns about differential anticipation effects, spillovers between states, or differences in shock responses related to petroleum exposure.

5.2 Baseline Continuous DID Estimates

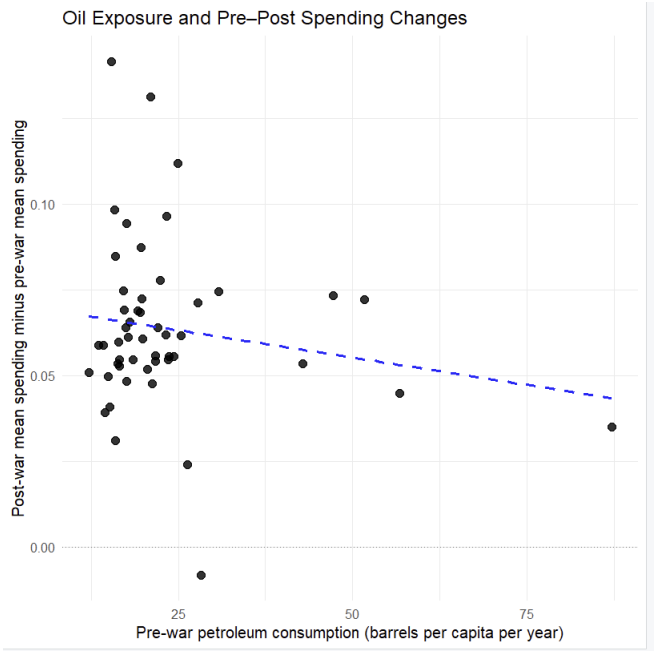


Figure 9 Pre-war Petroleum Exposure and Changes in Non-durable Spending

Figure 9 shows the relationship between pre-war petroleum exposure and changes in the non-durable spending indicator across U.S. states. Each point represents one state. The horizontal axis shows average petroleum products consumption per capita from 2019 to 2021, while the vertical axis shows the difference between the average spending indicator in the 27 post-war weeks and the 27 pre-war weeks.

Most points are above zero, indicating that the average nominal spending indicator was higher after the war than before the war in most states. However, the blue fitted line slopes downward, suggesting that states with higher pre-war petroleum exposure generally experienced smaller increases in the spending indicator.

This descriptive relationship is consistent with our expected negative relationship. However, the points are relatively widely spread around the fitted line, and there are fewer observations at higher exposure levels. Therefore, the fitted relationship may be sensitive to a small number of highly exposed states.

The baseline continuous DID regression gives an estimated coefficient of −0.004152 for the interaction between standardized pre-war exposure and the post-war indicator, with a state-clustered standard error of 0.002087.

This means that a one-standard-deviation increase in pre-war petroleum exposure is associated with a 0.00415-unit lower change in the spending indicator after the war relative to the pre-war period, equivalent to approximately 0.415 percentage points.

This estimate measures differences in a spending indicator defined relative to the common 2020 baseline. It does not mean that spending in dollars fell by 0.415%, nor does it imply that spending in highly exposed states necessarily declined in absolute terms. For example, even if spending increased across all states, states with higher petroleum exposure could still experience relatively smaller increases.

The estimated interaction coefficient has a p-value of 0.0523. It is statistically significant at the 10% level but does not reach the 5% significance level. Therefore, although the negative coefficient is consistent with our research hypothesis, the statistical evidence remains limited and is not strong enough to support a firm conclusion. The results suggest that higher pre-war petroleum-use intensity is associated with relatively weaker changes in nominal non-durable spending after the war.

```
OLS estimation, Dep. Var.: spend_nondurables_mean
Observations: 2,700
Fixed-effects: statefips: 50,  week_end: 54
Standard-errors: Clustered (statefips)
                  Estimate Std. Error  t value Pr(>|t|)
exposure_std:post -0.004152   0.002087 -1.98922 0.052275 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.031147     Adj. R2: 0.850352
                  Within R2: 0.004334
>
```

Since week fixed effects absorb changes common to all states, the estimated coefficient reflects differences in spending adjustments across exposure levels rather than the overall effect of the war on U.S. consumer spending.

Furthermore, as discussed in Section 4.3, the estimates may still be affected by differential anticipation effects, other state-level economic shocks, spillovers between states, and differences in shock responses associated with petroleum exposure.

Therefore, we interpret the baseline estimate as a relative spending change associated with pre-war petroleum exposure. The following section examines whether this relationship remains stable under alternative exposure measures and model specifications.

5.3 Robustness and Sensitivity Analyses

Variable / Statistic	(1) Baseline	(2) 2019 exposure	(3) Baseline + gasoline	(4) 2019 + gasoline
Exposure × Post	−0.0042* (0.0021)	—	−0.0041* (0.0021)	—
2019 exposure × Post	—	−0.0050* (0.0025)	—	−0.0050* (0.0025)
Gasoline price	—	—	0.0015 (0.0153)	0.0015 (0.0153)
State fixed effects	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes
SE clustering	State	State	State	State
Observations	2,700	2,700	2,700	2,700
R^2	0.85606	0.85624	0.85607	0.85625
Within R^2	0.00433	0.00559	0.00436	0.00561

To examine whether the baseline results are sensitive to the choice of exposure measure and control variables, we conduct three additional analyses: using petroleum exposure from 2019 only, adding

regional gasoline prices, and combining both adjustments. All models use the same sample of 50 states and 54 weeks, giving 2,700 observations. They include state and week fixed effects, with standard errors clustered at the state level. The 2019 exposure measure is scaled using the same exposure standard deviation as the baseline model to keep the coefficients comparable.

$$S_{it} = \alpha_i + \lambda_t + \beta_1(Exposure_i^{2019} \times Post_t) + \varepsilon_{it}$$

First, we replace the average petroleum consumption per capita from 2019 to 2021 with the value from 2019 only. This avoids directly including petroleum consumption changes during the COVID-19 pandemic in the construction of the exposure measure.

The estimated interaction coefficient is −0.004992, with a clustered standard error of 0.002489 and a p-value of 0. 050406. Compared with the baseline coefficient of −0.004152, the estimated effect remains negative and becomes slightly larger in absolute value. It is still statistically significant at the 10% level but does not reach the 5% level. This suggests that the negative estimated relationship does not depend entirely on including 2020 and 2021 in the exposure measure.

$$S_{it} = \alpha_i + \lambda_t + \beta_2(Exposure_i^{2019-2021} \times Post_t) + \gamma_2 GasPrice_{it} + \varepsilon_{it}$$

Second, we add weekly retail gasoline prices for each state's Petroleum Administration for Defense Districts (PADD) region to the baseline model.

Pre-war petroleum exposure measures differences in states' existing petroleum-use intensity, while gasoline prices describe the fuel price environment during the study period. Gasoline prices may affect other household spending through changes in transportation costs and may also directly affect the non-durable spending indicator, which includes purchases at gasoline stations. In addition, state and week fixed effects cannot fully account for changes in gasoline prices that differ across regions over time. Adding this variable therefore allows us to examine whether the estimated exposure coefficient remains stable after accounting for these regional price differences. The estimated interaction coefficient is −0.004139, with a clustered standard error of 0.002125 and a p-value of 0.057179. Compared with the baseline model, the coefficient changes very little, while the standard error increases slightly. The conclusion about statistical significance remains the same. The estimated coefficient on gasoline prices is 0.001539, with a p-value of 0.920441, indicating no statistically significant conditional relationship in this specification. Therefore, including regional gasoline prices does not substantially change the main estimate.

$$S_{it} = \alpha_i + \lambda_t + \beta_3(Exposure_i^{2019} \times Post_t) + \gamma_3 GasPrice_{it} + \varepsilon_{it}$$

Finally, we combine the 2019 exposure measure with the regional gasoline price control. The estimated interaction coefficient is −0.004980, with a clustered standard error of 0.002535 and a p-value of 0.055179.

This result is very close to the estimate of −0.004992 obtained using the 2019 exposure measure alone. Again, the coefficient is statistically significant at the 10% level but not at the 5% level. The gasoline price coefficient also remains statistically insignificant, with a p-value of 0.922919.

Overall, the estimated interaction coefficients are negative and similar in magnitude across all three additional specifications. This suggests that the relationship between higher pre-war petroleum exposure and relatively weaker post-war spending changes is reasonably stable under the alternative specifications considered.However, the p-values in all models remain slightly above 0.05.

Therefore, the findings should be described as having a consistent direction and similar magnitude but limited statistical support. They should not be interpreted as strong evidence of a causal effect.

6. Discussion and Limitations

6.1 Economic Interpretation of the Findings

Rising energy costs may have placed greater pressure on household budgets in states with higher petroleum-use intensity. Higher fuel prices increase necessary transport expenses, leaving less income available for other purchases. Businesses may also pass higher production and transportation costs on to consumers. If nominal income does not keep pace, households may reduce purchases or switch to cheaper alternatives. These channels provide possible explanations for the estimated relationship, but they are not directly tested in this study.

The nominal spending measure also allows competing effects to offset one another. Because the indicator includes gasoline station purchases, higher fuel prices may increase recorded fuel expenditure even as households reduce purchases elsewhere. Meanwhile, states with substantial petroleum production may benefit from higher industry income and employment. The estimated relationship may therefore reflect several offsetting channels and cannot directly establish changes in real consumption or household welfare.

The limited change in the exposure coefficient after controlling for regional gasoline prices does not establish that gasoline price transmission is unimportant. Week fixed effects absorb common national price movements, while PADD-level prices capture only regional variation and may not reflect the fuel costs faced by individual households. Gasoline prices may also lie on the transmission pathway of the shock. This specification should therefore be interpreted as a sensitivity analysis rather than a direct test of the mechanism.

6.2 Identification and Measurement Limitations

First, the findings depend on identification assumptions that cannot be fully verified. Although the pre-war Stute test does not reject the null hypothesis, it cannot establish that counterfactual parallel trends would hold after the invasion. Anticipation effects, differences in post-pandemic recovery, and other simultaneous shocks may also be related to petroleum exposure and spending changes. Moreover, states may respond differently because of industrial structure, transport options and government support. The estimated coefficient should therefore be interpreted as an exposure-related difference in spending adjustments rather than the causal effect of increasing petroleum exposure itself.

Second, both the exposure measure and the outcome have measurement limitations. State-level petroleum consumption includes household and business use, so it does not directly measure household fuel dependence. Although using 2019 exposure produces similar estimates, this does not eliminate concerns about pandemic-related changes. The Affinity indicator measures nominal spending and includes gasoline station purchases. Consequently, it cannot separate price changes from changes in quantities purchased or directly establish effects on real consumption and household welfare.

Third, the right-skewed exposure distribution and limited statistical precision warrant caution. A small number of highly exposed states, particularly Louisiana, may exert considerable influence on the estimated linear relationship. Leave-one-state-out analysis would help assess this sensitivity. The estimates also remain imprecise across specifications, limiting the strength of the conclusions.

Finally, the model captures average relative spending adjustments within a 27-week post-war window, rather than the full evolution or long-term effects of the shock. Trade, commuting and supply-chain connections may transmit effects across state borders, while state-clustered standard errors do not fully account for cross-state error correlation. The findings therefore describe short-term differences across exposure levels and cannot identify the overall effect of the war on U.S. consumer spending.

7. Conclusion

This study examines whether U.S. states with higher pre-war petroleum exposure experienced different changes in nominal non-durable spending following Russia's invasion of Ukraine. Using a continuous-exposure DID model with state and week fixed effects, we find that a one-standard-deviation increase in exposure is associated with a 0.415 percentage point smaller post-war increase in the spending indicator relative to the pre-war period. The evidence is suggestive rather than conclusive, as the baseline p-value is 0.0523.

The pre-war Stute test does not detect systematic exposure-related differences in spending changes, although this does not establish counterfactual parallel trends. Estimates remain similar in direction and magnitude when using 2019 exposure and controlling for regional gasoline prices. These findings suggest that aggregate spending increases may conceal differences in adjustments across states with different petroleum-use intensity.

The study contributes evidence on regional spending heterogeneity during the 2022 energy-price episode. However, the estimates do not establish the causal effect of increasing petroleum exposure or directly measure changes in real consumption and welfare. Future research could separate energy from non-energy spending and assess sensitivity to highly exposed states and alternative observation windows.

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